Olek Osikowicz

amosikowicz1@sheffield.ac.uk | olek-osikowicz.github.io | olek-osikowicz | olek-osikowicz

Education

PhD **University of Sheffield**, School of Computer Science — Sheffield, UK — Sept 2023 – present
- Efficient and reliable simulation-based Autonomous Driving Systems testing
- *PhD Supervisors: Donghwan Shin & Phil McMinn*

BSc **University of Sheffield**, Computer Science — Sheffield, UK — Sept 2020 – June 2023
- Graduated with **First-Class Honours**
- Dissertation: *Grounded In Reality Autonomous Driving Systems Testing*

Current Projects

Distributed Data Infrastructure
- Designed and deployed a multi-node computing cluster using Ray and AWS to parallelize software-in-the-loop evaluations
- Significantly improved the throughput of ML evaluation experiments in testing research group

Multi-Fidelity Test Generation
- Designed Multi-Fidelity Bayesian Optimization algorithms for Autonomous Driving Systems (ADS) testing
- Reduced costs of driving models evaluation by 16.8% compared to state-of-the-art baselines

Generative Modelling for Driving Scenarios
- Implemented a Variational Auto-encoder (VAE) model to learn compact latent representations of driving scenarios
- Designed, trained and evaluated the model enabling efficient sampling of driving scenarios for ADS testing

Flaky Test Analysis for ADS
- Empirically discovered and analysed the causes of flaky tests in simulation-based ADS testing
- Reported and published mitigation guidelines to reduce the impact of flaky tests on ADS verification

Employment

University of Sheffield, Research Assistant in Simulation-Based Testing — Sheffield, UK — June 2025 – present
- Developing automated Python tooling for large-scale ADS simulation and testing
- Integrated physics-based simulator CARLA with AutoWare for closed-loop evaluation

Dover Fueling Solutions, Summer Intern — Kraków, Poland — June 2022 – Sept 2022
- Built and validated automated ETL data pipelines on Microsoft Azure
- Worked with SQL warehouses and Databricks for scalable data processing

Skills

Programming: Python (advanced), C++, SQL, TypeScript, React, Svelte

Distributed computing: Ray, Docker, AWS (EC2/S3), GCP, SLURM, Multiprocessing, with Loki, Prometheus, Grafana

Machine Learning: PyTorch, Generative Models (VAEs), Bayesian Optimization, Reinforcement Learning

Publications

Multi-Fidelity Bayesian Optimization for Simulation-Based Autonomous Driving Systems Testing — Oct 2025
Olek Osikowicz, Phil McMinn, Wei Xing, Donghwan Shin
2026 IEEE Intelligent Vehicles Symposium (IV 2026)

Empirically Evaluating Flaky Tests for Autonomous Driving Systems in Simulated Environments — Apr 2025
Olek Osikowicz, Phil McMinn, Donghwan Shin
10.1109/FTW66604.2025.00009 *2025 IEEE/ACM International Flaky Tests Workshop (FTW 2025)*